WEEK 5

Omnichannel Optimization, Next Best Action, and Patient Predictive Modeling

One decision per prescriber per week: call, email, digital, or nothing.
How an NBA engine makes it — and how you know it worked.

Andi Cupallari · Applied Data Science and AI in Pharma Commercial · Fall 2026 · Monday, October 12

AGENDA

Tonight

2.5 hours · one notebook in two halves · one exercise · Homework 1 due tonight

0 **The question, and the trap**
Response vs. uplift — 15 min

1 **How an NBA engine works**
Three components, one loop, one design choice — 15 min

2 **Live 1 — the data, the trap, affinity prediction**
Notebook §1–4 — 35 min

3 **Live 2 — causal inference and uplift modeling**
Potential outcomes · confounding · S/T/X-learners · notebook §5 — 35 min

4 **Saturation, rules, scoring**
Notebook §6–7 — 15 min

5 **Evaluating a policy: OPE and bandits**
Notebook §8–9 — 15 min

6 **Patients, orchestration, exercise**
Notebook §10–11 — 20 min

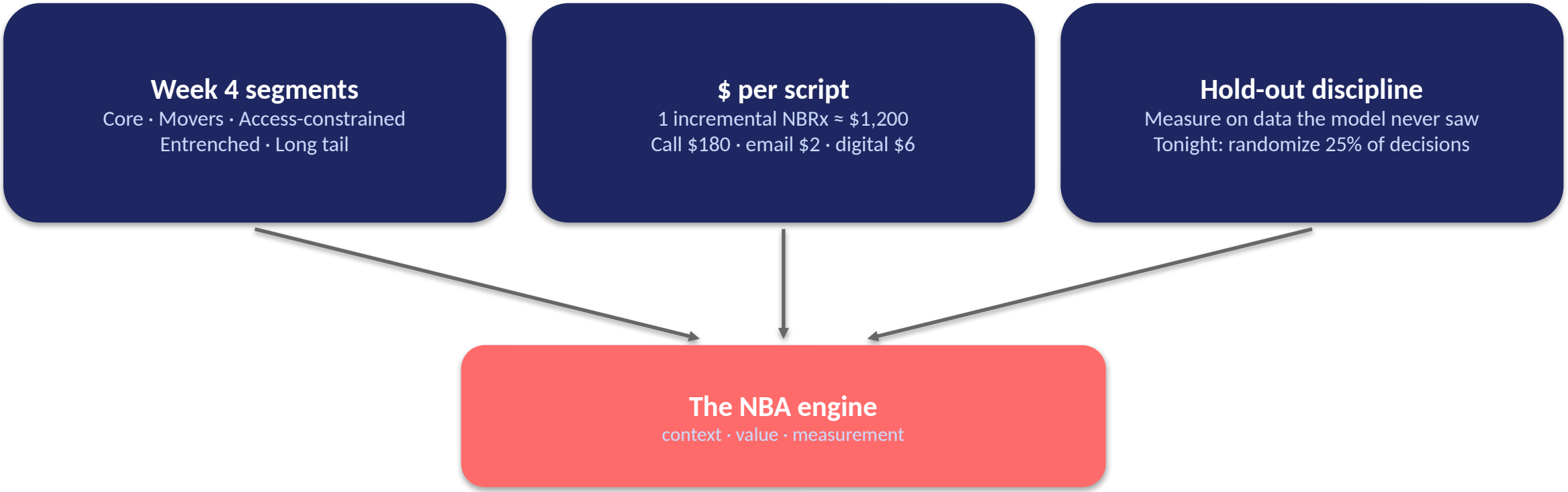
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The question, and the trap

Week 4 said who the prescribers are. Tonight: what to do with each one, this week.

0 · FROM WEEKS 3–4

What tonight inherits



Break-even for a $180 call: 0.15 incremental NBRx in two weeks. Most calls don't clear it.

0 · THE QUESTION

The Omnichannel Lead, every Monday

GLUCOREN · 6,000 targeted HCPs · four possible actions

“For each prescriber, this week: a rep call, an approved email, a digital placement, or nothing — and how will we know it worked?”

Rep call	Approved email	Digital placement	Nothing
$180 per touch	$2 per touch	$6 per touch	$0 per touch

6,000 HCPs × 26 weeks = 156,000 of these decisions in H1 2026

We owe back

- One action per HCP per week, with the expected incremental value behind it — and 'nothing' as a real answer

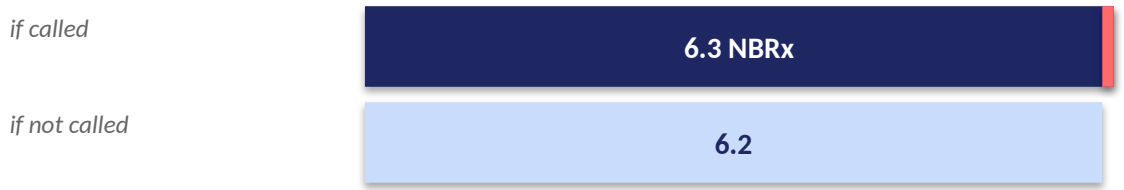
We will not pretend

- That 'who prescribes after a call' tells us whom to call — it tells us who prescribes

Response vs. uplift

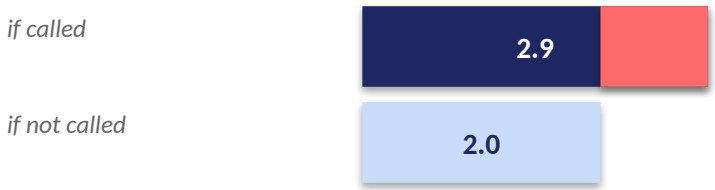
Response = what happens after the action. Uplift = that, minus what would have happened anyway.

Dr. Hale — Core



uplift +0.1 → −$60 net. A sure thing.

Dr. Bakshi — Movers



uplift +0.9 → +$900 net. A persuadable.

	prescribes anyway	wouldn't anyway
prescribes if touched	**Sure things** Core · uplift ≈ 0	**Persuadables** Movers · worth the touch
doesn't if touched	**Sleeping dogs** over-touched Entrenched · uplift < 0	**Lost causes** Long tail · nothing moves them

Why a response model fails

- It ranks Dr. Hale first — he prescribes most.
- Only persuadables return the $180.
- Finding them needs the difference, and the difference needs randomization.

Model the difference, not the level. The four types: Radcliffe & Surry (2011).

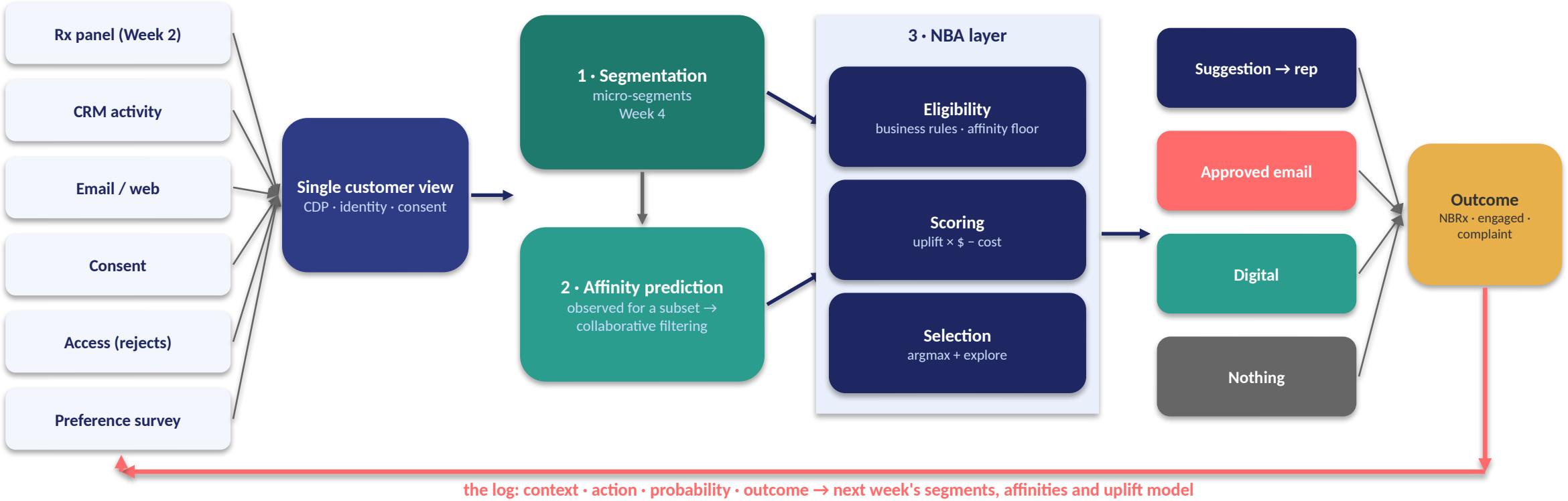
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How an NBA engine works

Three components — segmentation, affinity, NBA — one weekly loop, one design choice that makes it learnable.

An NBA engine in one picture

Three components: segmentation → affinity → NBA. Terms: suggestions (Veeva), orchestration (IQVIA), business rules (Axtria), single customer view (McKinsey)



Segmentation says who. Affinity says how to reach them. The NBA layer says whether it pays, under the rules. The log feeds all three.

1 · THREE COMPONENTS

Segmentation → affinity → NBA

What each answers, what it needs, where tonight builds it

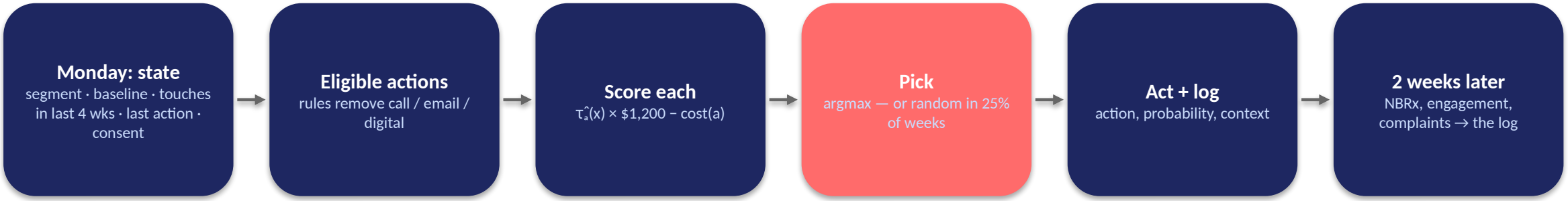
	1 · Segmentation / micro-segmentation	2 · Affinity prediction	3 · NBA layer
Question	Who is this HCP?	Which channels reach them?	Which action pays this week?
Input	Rx, patient mix, switching, channel response (Week 4)	Observed engagement on a subset of HCP × channel cells; a preference survey	Randomized touch log; rules; affinities; value per script
Method	k-means, two-stage; supervised growth score	Collaborative filtering (matrix factorization) + segment prior for cold start	Uplift model (X-learner); business rules; argmax with exploration
Output	Segment and growth group per HCP	Affinity per HCP per channel (full matrix)	One action per HCP per week, with its probability
Fails when	Unstable across refreshes	Cold start with no prior; stated ≠ revealed	No randomization; engagement used as the outcome
Tonight	Inherited from Week 4	Notebook §4	Notebook §5–8

IQVIA's NBA engine: 'dynamic micro-segmentation and AI/ML-based scoring' on a '360-degree view of HCP preferences' — the same three parts.

1 · THE LOOP

One HCP, one week

The same cycle, 6,000 times every Monday



Example — Dr. Bakshi (Movers), week of Mar 2

Monday: state	Eligible actions	Score each	Pick	Act + log	2 weeks later
Movers · 2.0 NBRx/2w · 1 touch · last: none · consent ✓	call ✓ email ✓ digital ✓ none ✓	call +0.90 → +$900 · email +0.30 → +$358 · digital +0.45 → +$534	call	logged: call, p = 0.62	2.9 NBRx · call completed

Next week the state has changed (2 touches, last action = call) — and so does the decision.

Randomize a share of the decisions

Industry: a test-and-control design (ZS); McKinsey: 'A/B testing in the field'. Bandit language: the exploration arm.

156,000 HCP-weeks

75% — legacy rules engine
Call frequency by decile · fortnightly email to consented HCPs · digital on rotation
Who gets called depends on decile — the same thing that drives prescribing

25% — randomized
none · call · email · digital, each with probability 0.25
regardless of who the HCP is

Learns the targeting rule

- Any model fit here learns 'high decile → call → high NBRx'. Useful for affinity (§4); useless for uplift.

Learns the effect

- Uplift (§5), the fatigue curve (§6), OPE (§8) all come from here.

The vocabulary, and where it comes from

Definitions in the Field Guide (Part 11); sources in the Reading Pack (Part F)

Term	Who uses it this way
Next Best Action (NBA)	IQVIA 2025; Veeva Systems 2015; EY 2023; McKinsey 2022
Suggestions / triggers / signals	Veeva Systems 2015; Axtria 2020; ZS 2024; IQVIA 2026
Next Best Experience (NBX)	ZS 2026; Axtria 2025
Omnichannel orchestration	IQVIA 2026; IQVIA 2024; Axtria 2025; McKinsey 2022
Micro-segmentation	IQVIA 2025; McKinsey 2022
Channel preference / affinity; affinity prediction (collaborative filtering)	IQVIA 2024; IQVIA 2018; IQVIA 2023; IQVIA 2025; Koren, Bell & Volinsky, IEEE Computer 2009
Saturation / fatigue; frequency cap	IQVIA 2018; IQVIA 2023; Axtria 2025
Business rules / suppression rules	Axtria 2025; IQVIA 2026
Promotion response	Axtria 2021; ZS 2024
Uplift / incremental lift; persuadables, sure things, lost causes, sleeping dogs	Stochastic Solutions (N. Radcliffe & P. Surry) 2011; Athey & Imbens, PNAS 2016
Test and control; holdout; randomized test cell (the 'exploration arm')	ZS 2026; McKinsey 2022; IQVIA 2026
Engagement vs. impact	ZS 2026; IQVIA 2025
Single customer view / customer 360 / common data spine / CDP	IQVIA 2026; IQVIA 2024; McKinsey 2022; EY 2023
Closed-loop learning; suggestion acceptance	ZS 2026; IQVIA 2025; Axtria 2025
Human in the loop; guardrails	McKinsey 2025; ZS 2026; IQVIA 2025
Patient journey; closed patients; provider behavior	Komodo Health 2023; IQVIA 2025
Contextual bandit; offline policy evaluation (IPS, SNIPS, doubly robust)	Li, Chu, Langford & Schapire (WWW); Dudík, Langford & Li (ICML) 2010

02

Live, part 1 — the data, the trap, affinity prediction

Notebook §1–4

One row = one HCP, one week, one decision

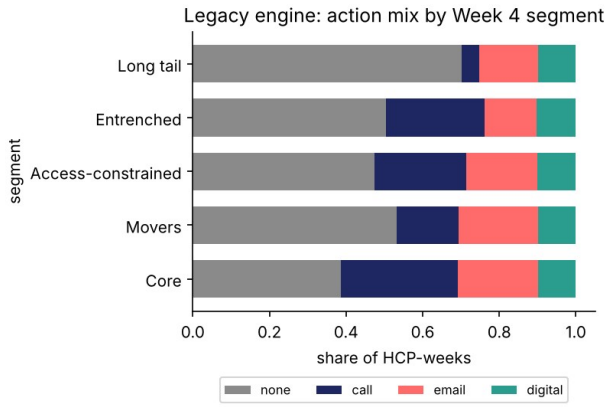
glucoren_touch_log.csv · 156,000 rows · plus the Week 4 segments file

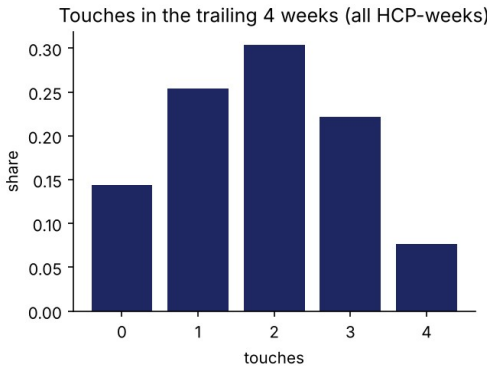
Context
segment · decile
trailing NBRx
calls / emails / digital in 4 wks
weeks since call · last action
consent · reject rate

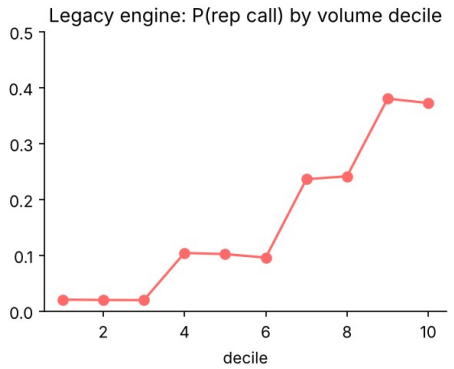
Action
none · call
email · digital

Probability
p(action | context)
0.25 if randomized
the rules' prob. otherwise

Outcome
NBRx next 2 weeks
engaged (open / click / completed)
complaint



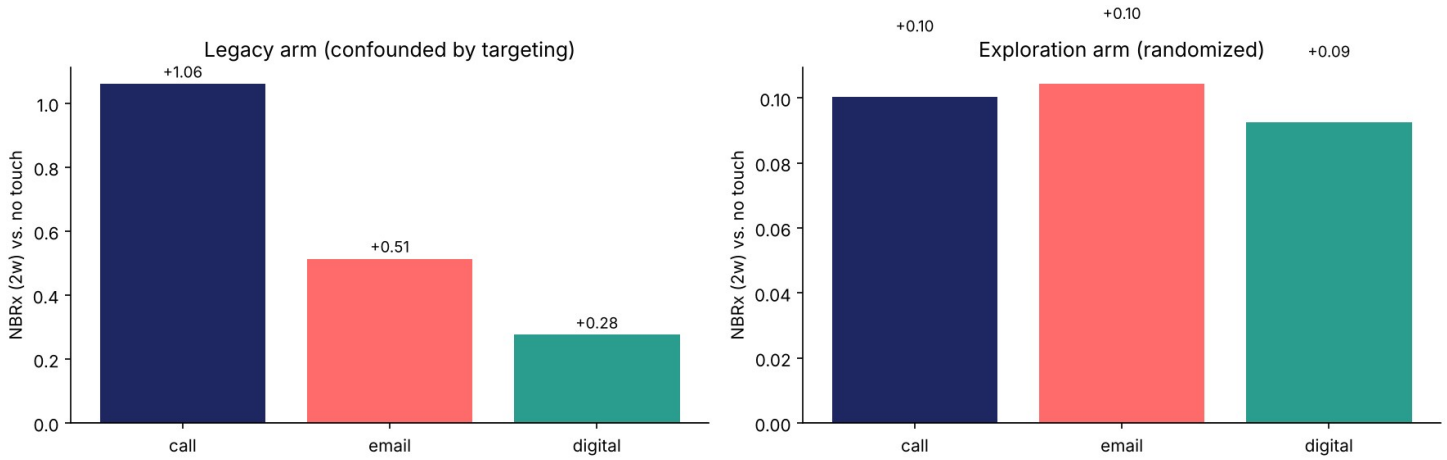




1.83
touches per HCP per 4 weeks (legacy)

2,450 → 135
emails without consent → complaints

Called vs. not called: +1.06 NBRx on the legacy arm, +0.10 randomized



Response model, top decile

- 62% Core · 30% Movers · 0% Entrenched
- It learned the call plan and called it response.

Honest average effect of a call

- +0.10 NBRx = $120 for $180. On average, a call loses money.

Not 'does a call work' but 'for whom does a call pay' — a heterogeneous-effect question.

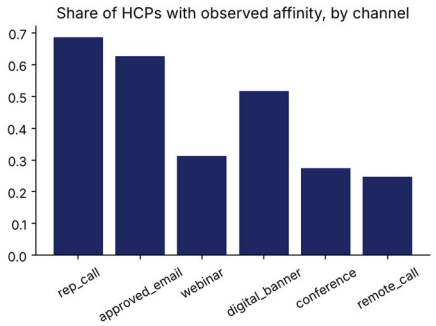
Affinity is known for a subset — 44% of the HCP × channel cells

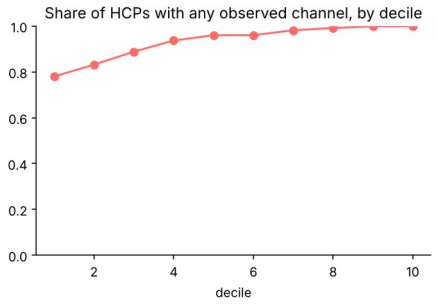
Six channels. History where the brand has touched the HCP on that channel; a 600-HCP preference survey covers all six.

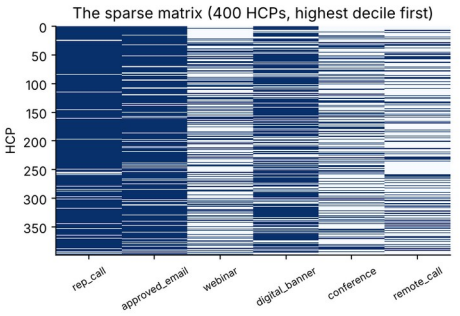
	rep call	email	webinar	digital	conf.	remote
Dr. Adler	.61	.33		.28		
Dr. Bakshi	.48	?		.55		
Dr. Costa	.36		?			
Dr. Dunn	?					?
Dr. Eng	.41	.24	.33	.27	.31	.35
Dr. Farah	.58	.30				
Dr. Goyal	?			.47		
Dr. Hale	.66	.31	.49	.29	.57	.40
...						

navy = observed engagement rate · ? = what the engine needs

Dr. Eng, Dr. Hale: surveyed (all six). Dr. Dunn: cold start.







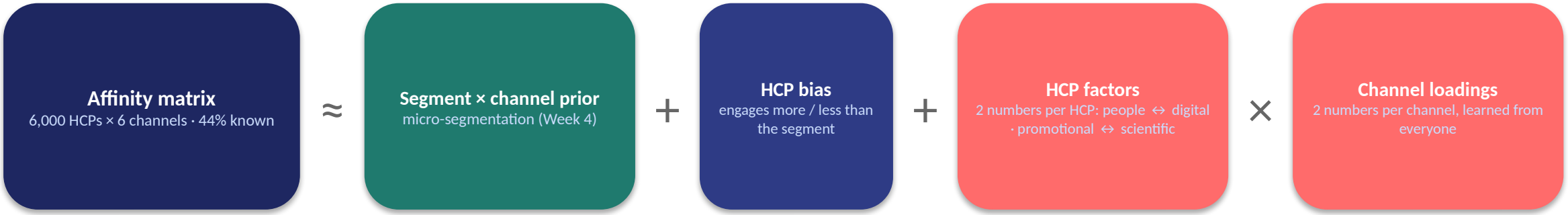
Why it is sparse, and sparse in a biased way

- Coverage follows the legacy call plan: high-decile HCPs are 'known' on call and email; webinar, conference and remote are rarely offered to anyone.
- 404 HCPs have no observed channel at all. The engine must still decide for them.

A sparse HCP × channel matrix with blanks to fill is a recommender-system problem.

Filling the matrix with collaborative filtering

Matrix factorization with biases (Koren et al., 2009) on the residual from the segment prior; segment prior alone for cold start



RMSE (lower is better)	Held-out cells	HCPs with ≥ 3 known cells	Cold-start HCPs
Segment × channel mean	0.184	0.182	0.185
kNN on Week 4 features	0.188	0.190	0.189
Matrix factorization	0.161	0.131	0.290
Hybrid (MF + prior) — ships	0.146	0.131	0.185

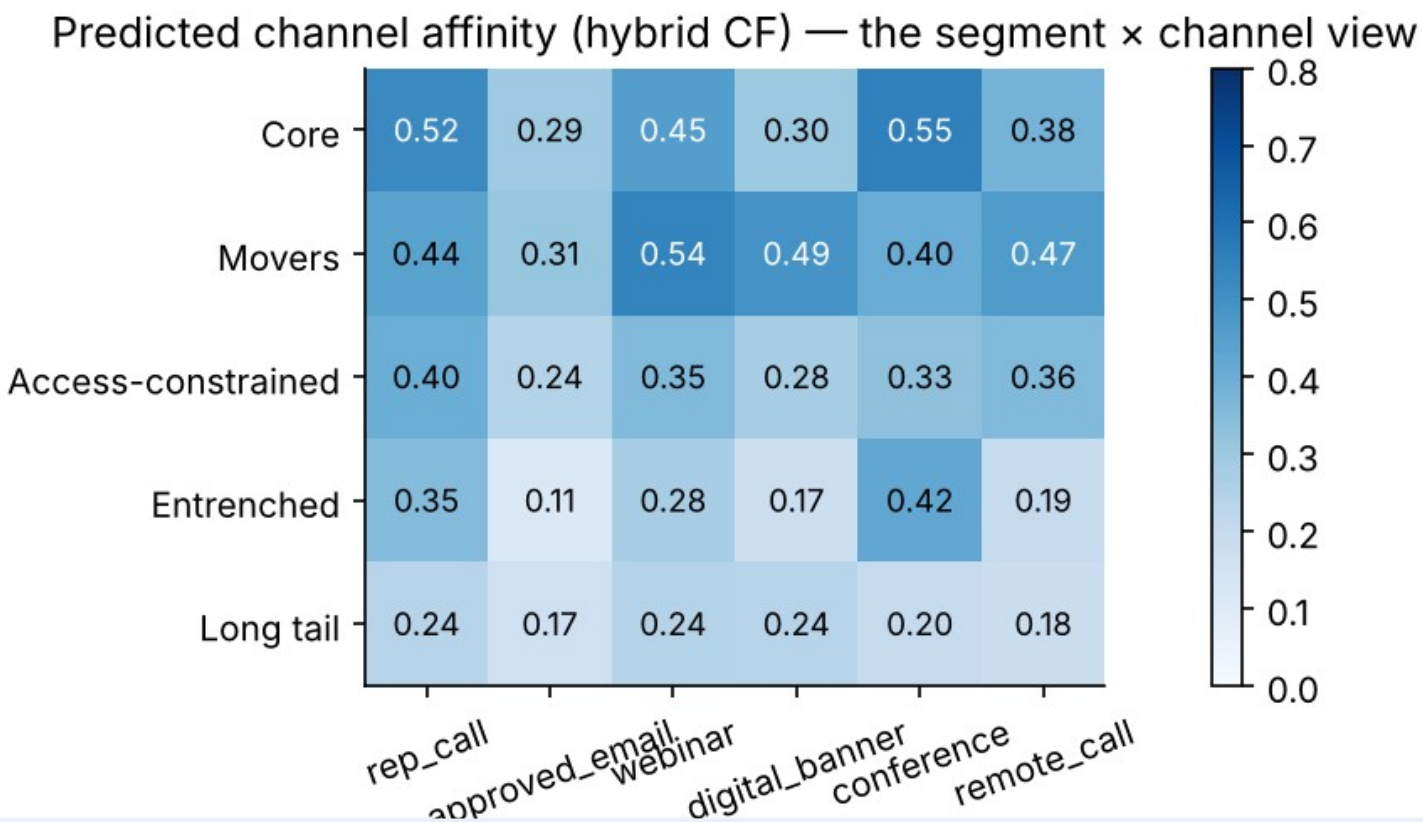
Read it

- The segment prior is a strong baseline — that is what a good micro-segmentation buys.
- CF improves with history: 2–3 known channels place the HCP in the latent space.
- Cold start: nothing to factorize → the prior. Hybrid ships.

Correlation with the planted full matrix on the 56% of cells nobody observed: prior 0.55 → hybrid 0.68.

The full matrix — and a check against revealed engagement

Hybrid CF output, averaged by segment. Validation: predicted affinity vs. observed engagement rate in the touch log, r = 0.20 / 0.29 / 0.33 (call / email / digital)



Affinity ≠ uplift

- Core: highest affinity on five of six channels, lowest return per dollar (§5).
- Movers: 0.49 digital, 0.54 webinar — affinity and uplift agree.

How the engine uses it

- Eligibility: a channel with predicted affinity < 0.15 is not offered (email blocked for 34% of HCPs, digital 25%, call 15%).
- Stated preference stays the other half: IQVIA Channel Dynamics.

Affinity decides how to reach an HCP. Uplift (§5) decides whether it pays. The segmentation sits under both.

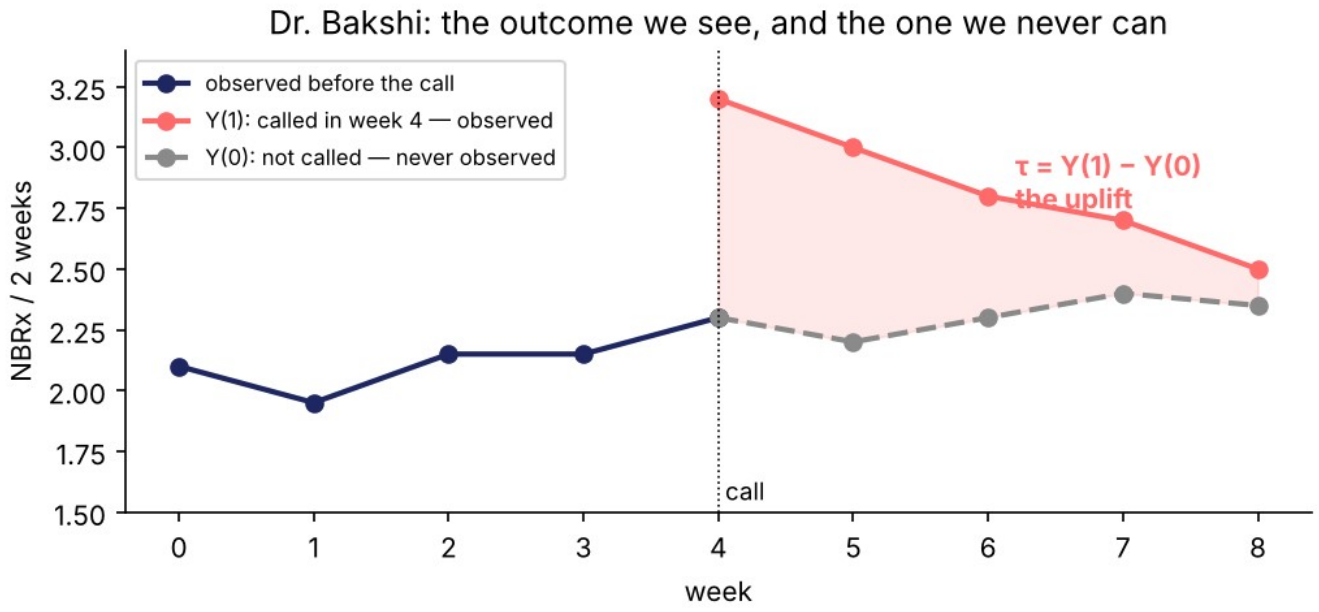
03

Uplift modeling — a short course in causal inference

Why 'who prescribes after a call' and 'what the call did' are different questions, and how S-, T- and X-learners answer the second. Notebook §5.

Two outcomes per HCP, one of them observed

The Neyman–Rubin framework: every HCP-week has an outcome if called, Y(1), and an outcome if not, Y(0)



$$\tau_i = Y_i(1) - Y_i(0)$$

the uplift of a call for HCP i

The fundamental problem

- We observe Y(1) or Y(0), never both. No row in any dataset has an uplift label.

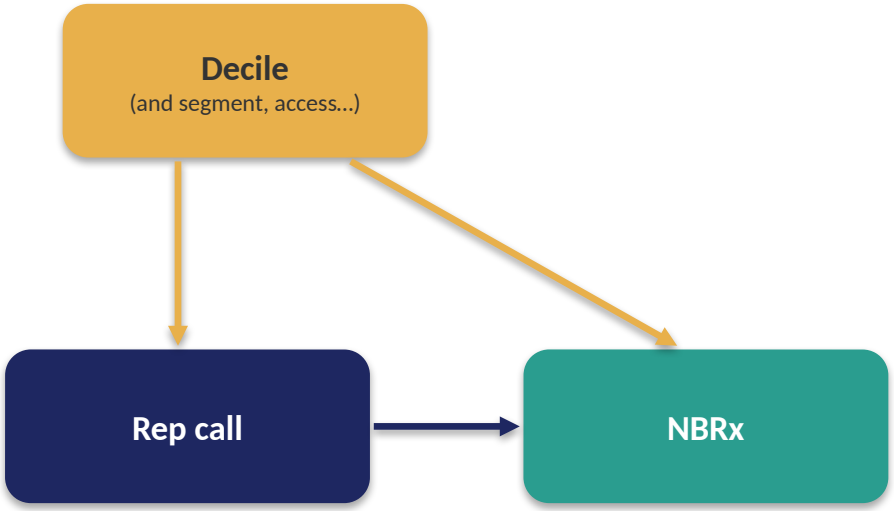
So we estimate averages

- ATE = E[Y(1) − Y(0)]: the average over everyone.
- CATE τ(x) = E[Y(1) − Y(0) | X = x]: the average for HCPs like this one. The engine needs τ(x).

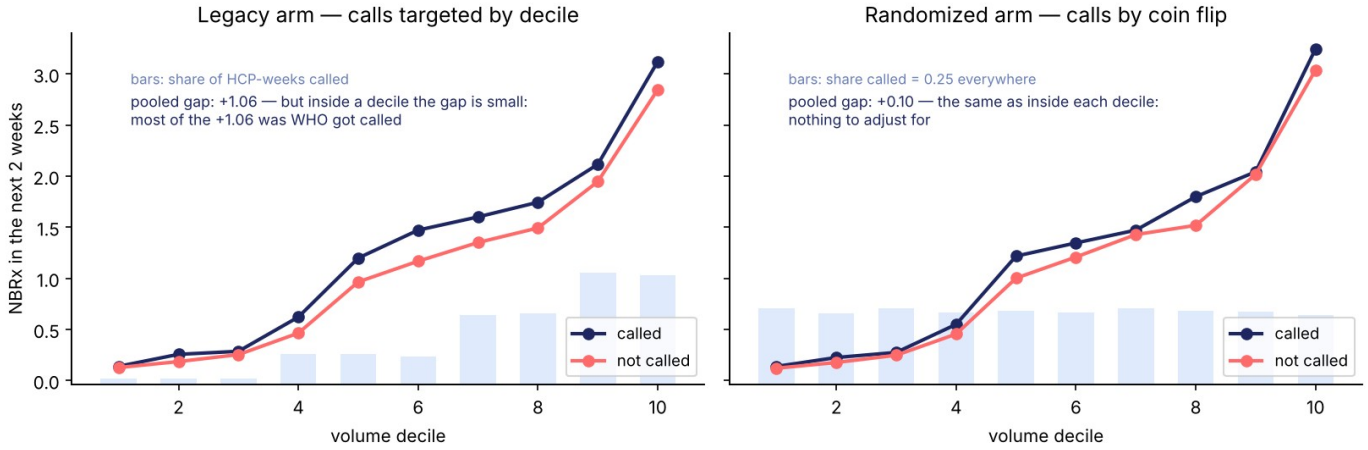
Uplift is a difference between two worlds. Only one is ever in the data.

Why the naive comparison lies

Called vs. not called = the effect of the call + whatever made the rep call in the first place



The confounder opens a back door: decile raises both the chance of a call and the NBRx. The naive gap mixes the navy arrow with the gold path.
Pooled gap +1.06; inside a decile it is small.



Two ways out

- Adjust for the confounders — works only if you know and measure all of them (here it is decile by construction; in reality it never is).
- Randomize the action — cuts the gold arrow into 'Rep call'. The exploration arm.

Observational: effect + selection. Randomized: effect. That is the whole case for the 25%.

What randomization buys: three conditions, satisfied by design

The conditions under which a difference in means is the effect (Rubin's assumptions, in pharma words)

Exchangeability

Called and not-called HCP-weeks are alike except for the call. Coin flip ⇒ true. Legacy arm ⇒ false (decile).

Overlap (positivity)

Every kind of HCP had a real chance of every action. 0.25 each ⇒ true. Legacy: Long-tail HCPs are almost never called ⇒ no overlap there.

Consistency / no interference

A 'call' means the same thing for everyone, and one HCP's call does not change another's outcome. Mostly true; peer effects are the exception.

ATE = E[Y | call] − E[Y | none] on the randomized rows = +0.10 NBRx per 2 weeks ($120 of margin for $180)

When you cannot randomize

- Propensity weighting, matching, difference-in-differences, synthetic controls (ZS 2026) — all rest on exchangeability you have to argue for.

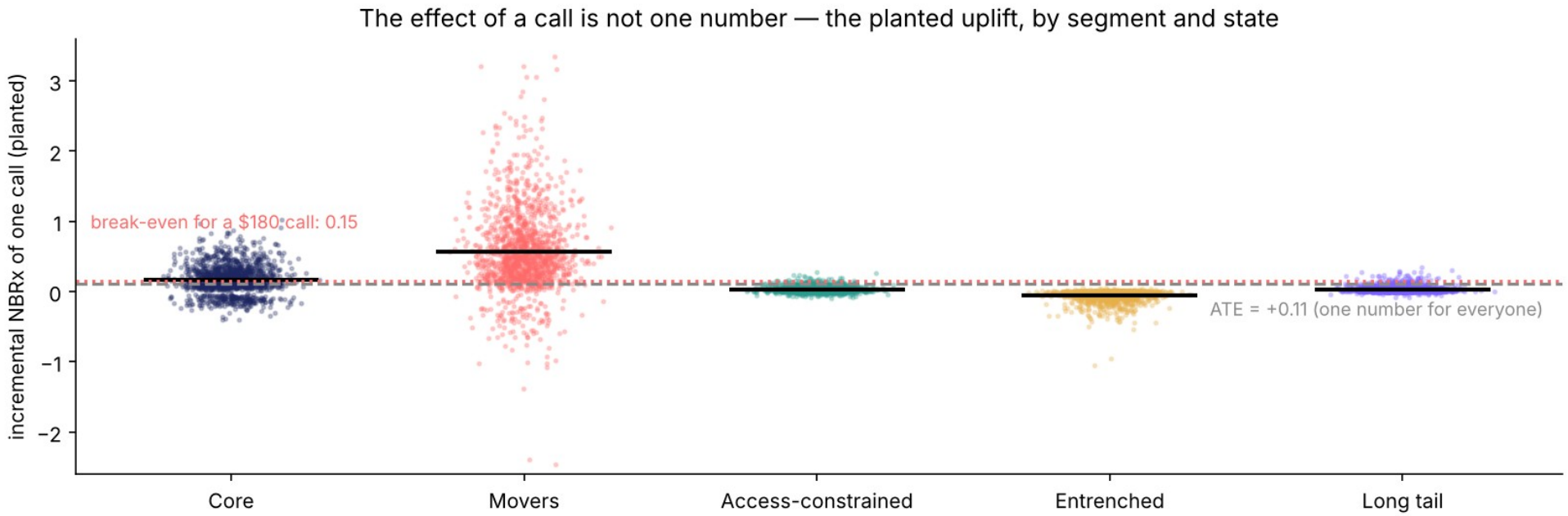
Why the exploration arm stays on

- Next quarter's CATE, OPE and the fatigue curve all need rows where these three hold.

Randomization is not a statistics nicety. It is what makes the number mean what the brand thinks it means.

One number is not enough

The planted uplift of a call, for every hold-out HCP-week, by Week 4 segment



The ATE hides the decision

▸ +0.11 on average; Movers +0.6, Entrenched below zero. A call pays for Movers and loses money almost everywhere else.

CATE τ(x) is the target

▸ x = segment, baseline, touches this month, last action, consent, access. The uplift model estimates τ(x) from the randomized rows.

The engine does not need the average effect. It needs the effect for this HCP, this week.

Same data, different target

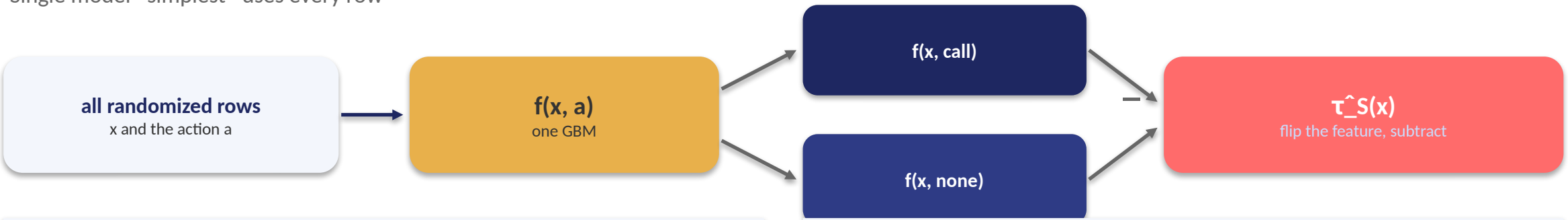
A response model and an uplift model can share features, algorithm and code — and answer different questions

	Response (prediction) model	Uplift (causal) model
Target	E[Y | x, call] — NBRx after a call	$\tau(x) = E[Y \mid x, call] - E[Y \mid x, none]$
Label in the data	Yes: NBRx	No: never observed for any row
Rows it can use	Any — but learns the targeting rule on observational rows	Randomized (or credibly adjusted) rows only
Ranks first	Sure things — Core, decile 10	Persuadables — Movers
Evaluated by	AUC, RMSE, R^2	Uplift by decile, Qini — on groups, not rows
Pharma name	Promotion response; 'propensity to prescribe'	Uplift / incremental lift; heterogeneous treatment effect

A high-AUC response model can rank HCPs by uplift no better than chance. Different target, different model.

S-learner: one model, the action is a feature

Single model · simplest · uses every row



Why it under-shoots

- The action is one feature among ~40. Regularization treats it like any other: where the tree does not split on it, $\hat{\tau} = 0$.
- Tonight: predicted sd 0.21 vs planted 0.31 — compressed toward zero; fewer HCPs cross break-even than should.
- Correlation with truth 0.74 (highest) — it gets the order right and the size wrong.

When to use it

- Small data; many actions (one model, not one per arm).
- When the effect is large relative to noise.
- Qini tonight (call / email / digital): 133 / 75 / 74.

Simple and honest about order; biased toward 'no effect'.

T-learner: one model per arm, then subtract

Two models · no shrinkage · noisy where an arm is thin



Why it over-shoots

- Each model sees only its own rows (≈ 7,000 each); the two fit the noise differently and the difference inherits both errors.
- Tonight: predicted sd 0.52 vs planted 0.31 — too spread; correlation 0.50.
- Digital Qini 18: a small effect drowned by two noisy models.

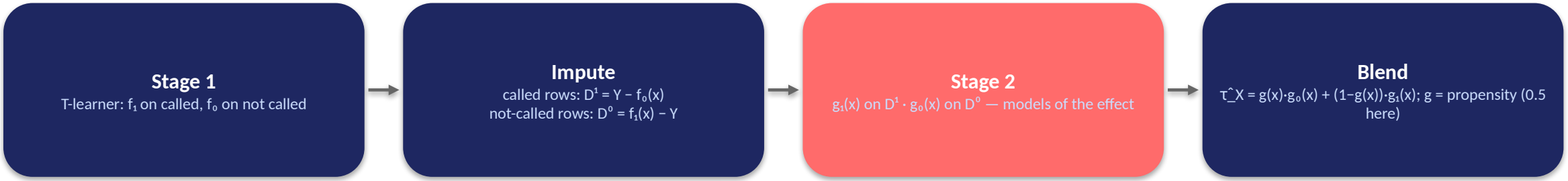
When to use it

- Arms of similar size with plenty of rows; outcomes that differ in shape by arm.
- Qini tonight: 174 / 51 / 18.
- The baseline every other learner is measured against.

No bias toward zero — but the difference of two noisy models is noisier still.

X-learner: impute each row's effect, then model the effect itself

Two stages · built for unequal arms · the production model tonight



Why the second stage helps

- Stage 2 fits a model of τ directly on pseudo-labels, so the effect is the target, not a leftover of two outcome fits.
- Each row is used twice — once for its own arm, once imputed — so thin arms borrow strength.
- Tonight: sd 0.42 vs 0.31; correlation 0.59; Qini 192 / 76 / 62 — best or tied on every channel.

When to use it

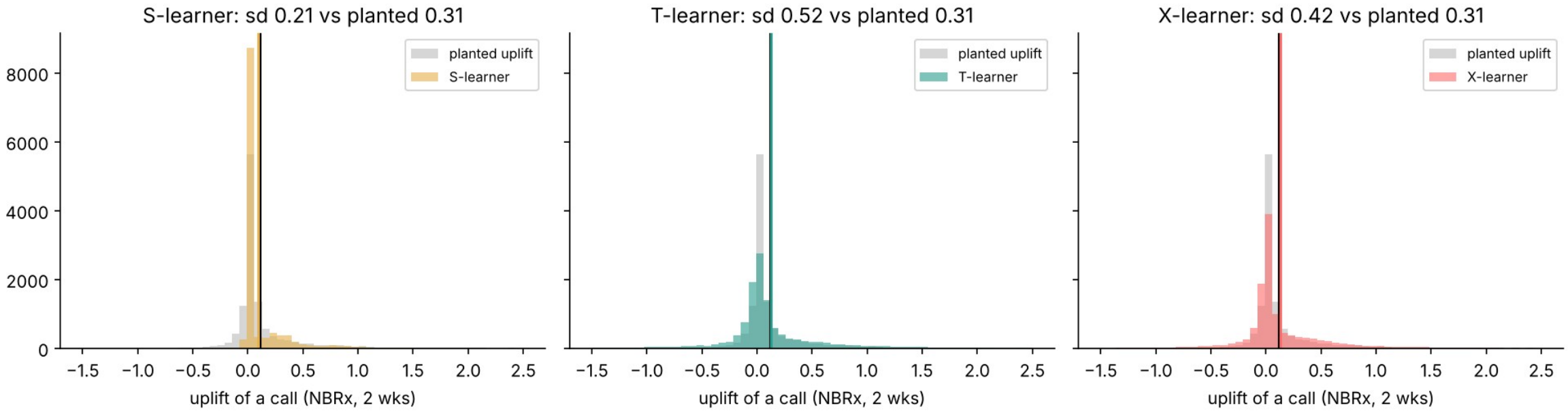
- Unequal arms — the normal case: a 90% control arm in a real log.
- Small effects in noisy outcomes (the email).
- The blend weight is the propensity; with a 25% exploration arm it is known, not estimated.

The effect becomes the target. That is why it wins on the email and ships in §7.

3 · WHAT EACH PRODUCED

The same hold-out, three distributions of τ̂

Planted uplift (grey) vs. each learner's predictions; vertical lines = means



S shrinks

▸ sd 0.21 < 0.31. The mass sits at zero.

T spreads

▸ sd 0.52 > 0.31. Long tails on both sides are noise.

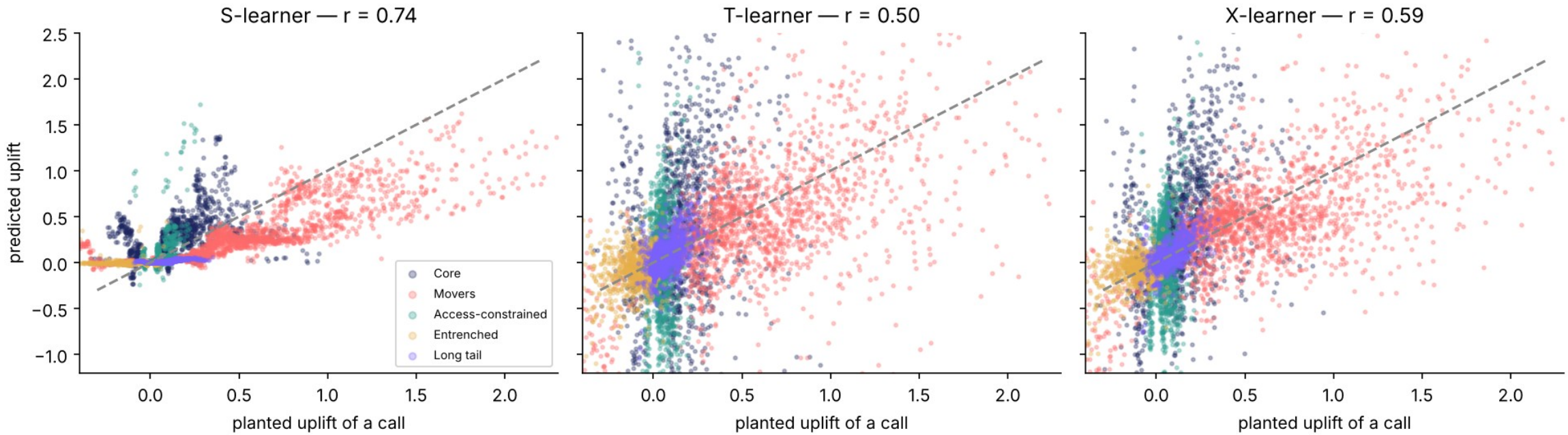
X is closest

▸ sd 0.42. Still too wide — 26 weeks of a 25% arm.

Three estimators of the same quantity disagree on its scale. The ranking is what the engine uses; the scale is what the finance case uses.

Predicted vs. planted uplift, hold-out HCP-weeks

Synthetic data's one luxury: the label that never exists in a real log



Read the colour

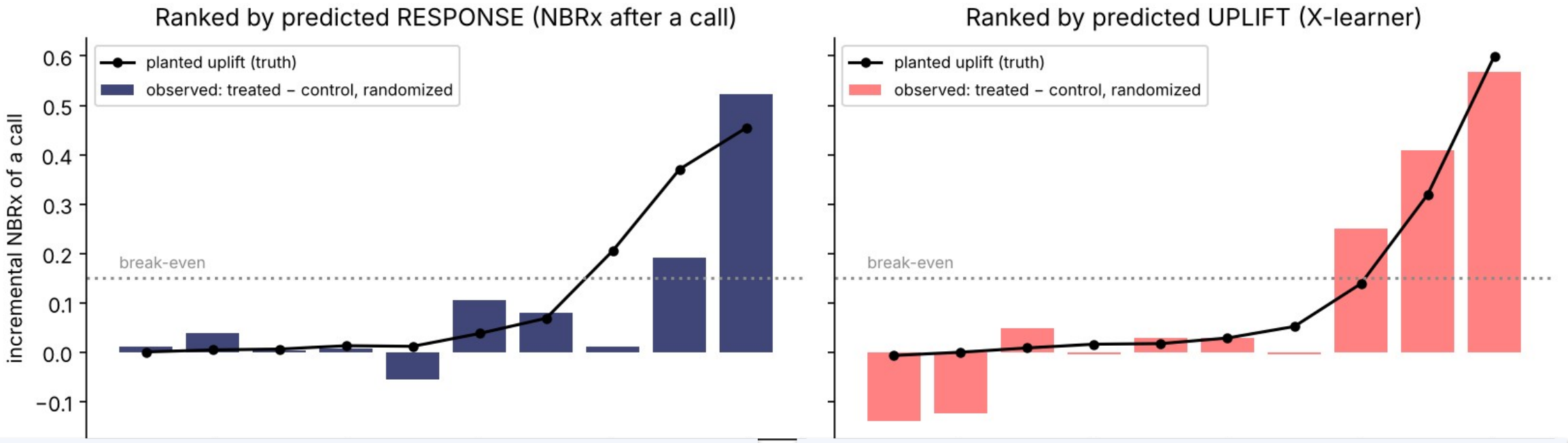
- Movers (coral) are the persuadables — every learner puts them top right. Entrenched (gold) sit at or below zero. Within a segment the spread is the state: touches this month, last action.

Segment order: easy. Within-segment state: harder. Individual HCP: beyond this data.

3 · THE PAYOFF

Rank by response, or rank by uplift — same HCPs, same randomized outcomes

Hold-out HCP-weeks cut into deciles of each ranking; bars = treated − control inside the decile (randomized, so this is the real effect)



Response ranking

▸ Top decile +0.45 — but decile 9 (Core) barely clears break-even and deciles 1–8 are flat: it found big baselines, not persuadables.

Uplift ranking

▸ Deciles 8–10 all above break-even and rising; the bottom deciles are sleeping dogs the engine should not touch.

The response model's top decile is worth calling. The uplift model's top three are — and it knows whom to leave alone.

Meta-learners side by side — and the causal forest

All on the 39,000 randomized HCP-weeks, hold-out split by HCP

S-learner

one model
f(x, action)

$\hat{\tau} = f(x, \text{call}) - f(x, \text{none})$

simple · shrinks the effect toward 0

T-learner

model on call rows

model on none rows

$\hat{\tau} = f_1(x) - f_0(x)$

honest · noisy where rows are few

X-learner

T-learner

impute each row's effect

model the effect itself

built for unequal arms · regularizes

Qini (hold-out)	S	T	X	response model
call	133	174	192	110
email	75	51	76	72
digital	74	18	62	33

Causal forest (optional)

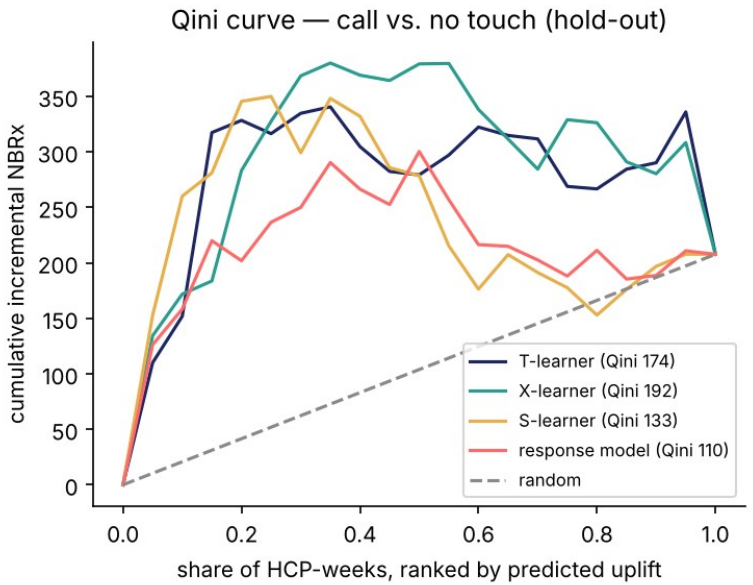
▸ econml CausalForestDML — honest trees on the effect (Athey & Imbens). Same data limits, valid intervals.

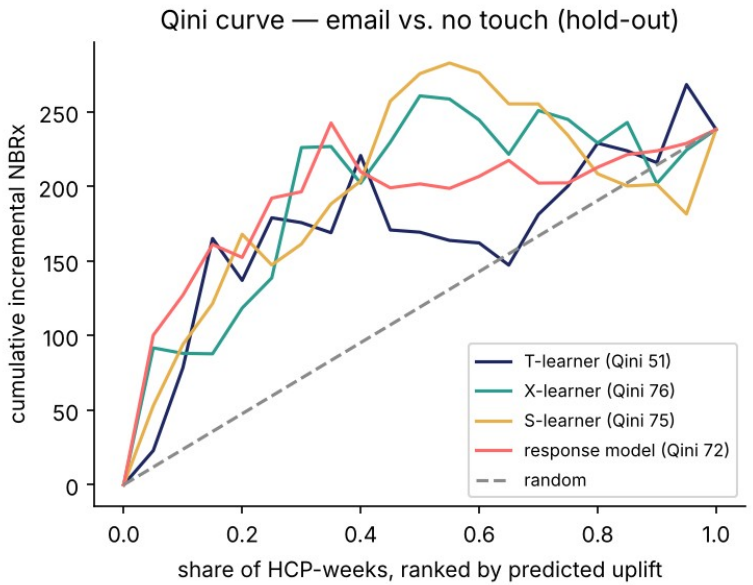
X-learner wins or ties every channel → it is the production model from §7 on. Grade on Qini, never on R^2.

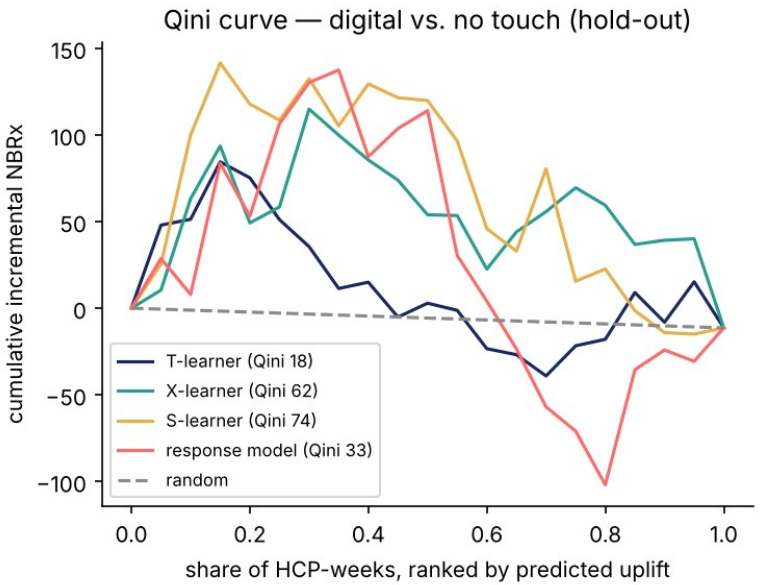
3 · QINI

Evaluating uplift without individual labels

Rank by predicted uplift → inside each decile, compare randomized treated vs. control → cumulate







Call

▸ Every learner beats the response model. X-learner top deciles: +0.60, +0.52 observed NBRx.

Email

▸ T 51 → X 76: the second stage finds a small effect in noisy counts.

A curve that rises fast then flattens found the persuadables and ran out.

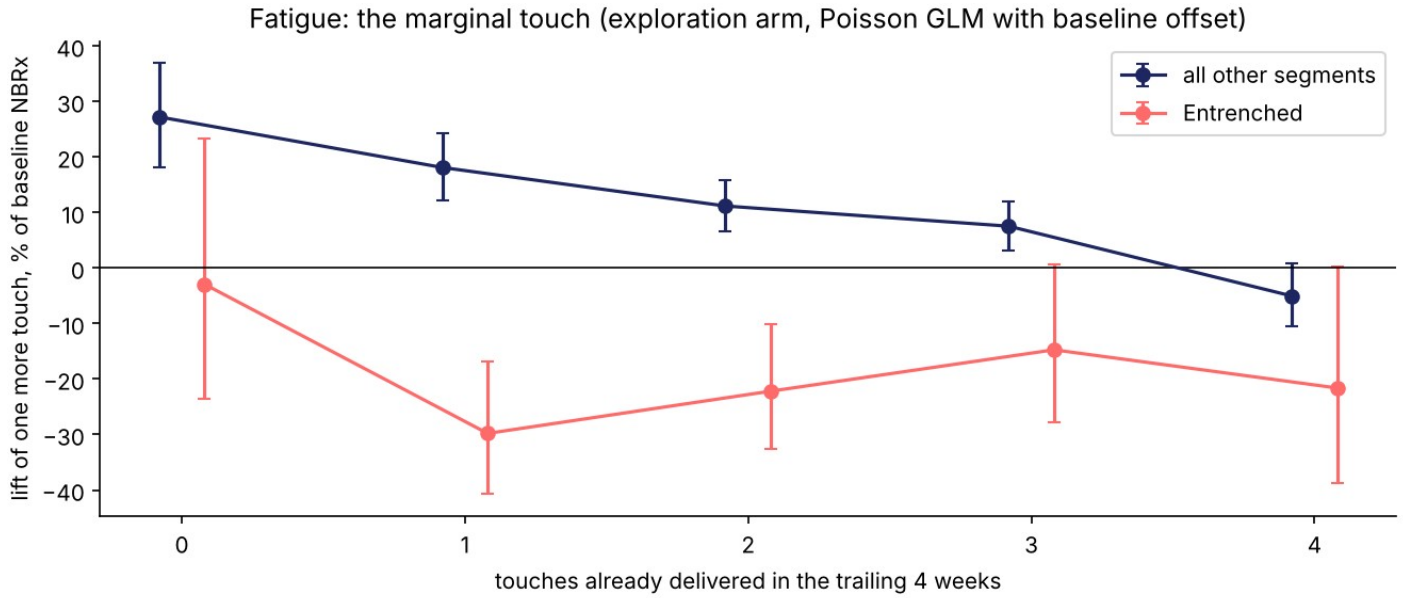
04

Saturation, rules, and the scoring layer

The marginal touch has its own curve. The rules fall out of it; the score sits behind them. Notebook §6–7.

The marginal touch

IQVIA (2018): digital's early success 'resulted in overkill and saturation'. Measured here on the randomized rows.



Touches already (4 wks)	Others	Entrenched
0	+27%	−3%
1	+18%	−30%
2	+11%	−22%
3	+8%	−15%
4	−5%	−22%

Two more rules

- Email after a quiet week +10%; after a call +2%.
- No consent: 2,450 emails, 135 complaints, no lift.

Cap: 3 touches per 4 weeks (2 for Entrenched). No email within a week of a call. Consent is a hard gate.

4 · BUSINESS RULES

Rules run before the model

Axtria: 'AI/ML models, business rules, and governance'

Rule	Definition	Type	Owner
Consent	No email without opt-in; no retargeting without cookie consent	Hard	Compliance
Frequency cap	≤ 3 touches / 4 weeks (≤ 2 Entrenched)	Hard	Brand + Field
Call recency	No call within 2 weeks of a call; field may override with a reason	Soft	Field
Sequencing	No email the week after a call	Soft	Omnichannel
Complaint hold	8 weeks of email silence	Hard	Compliance
MI hold	No promotion while a medical-information request is open	Hard	Medical
Access trigger	Reject rate > 25% → access content replaces efficacy content	Content	Market Access
Affinity floor	Predicted affinity < 0.15 → channel not offered (§4)	Hard	Analytics + Omnichannel
Holdout · exploration	10% of HCPs outside the engine · 25% of weeks randomized	Design	Analytics

Going into next week: cap blocks 42% of HCPs, consent 25%, recency 32%, affinity floor 50% (some channel). The engine's first job is to say no.

4 · SCORING

One HCP, four scores, one answer

value(a) = $\hat{\tau}_a(x)$ × $1,200 − cost(a) · Dr. Eng, Access-constrained, 1 touch this month, consent ✓, last action: none

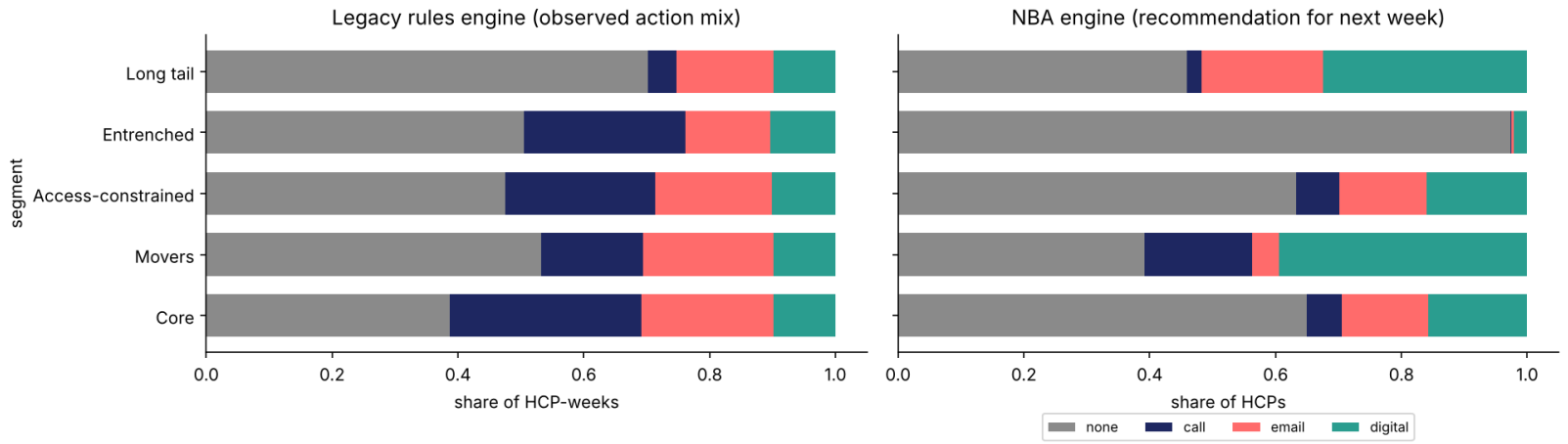
action	$\hat{\tau}$(NBRx, 2 wks)	× $1,200	− cost	= value	eligible?
Call	+0.05	$60	$180	−$120	× recency rule
Email	+0.35	$420	$2	+$418	✓
Digital	+0.08	$96	$6	+$90	✓
Nothing	0	$0	$0	$0	✓

argmax over eligible actions → approved email (access/copay content). Suggestion to the rep: none this week.

Across 6,000 HCPs: calls 955 → 297 a week; 60% get nothing. Model-estimated value $838k vs $169k for the legacy mix.

Legacy engine vs. NBA engine — next week's mix

Same model scoring both



What moved

- Calls: from Core and Entrenched to Movers.
- Movers → digital (39%).
- Access-constrained → the access email.
- Entrenched: 97% nothing.
- Top 10 by value: nine Movers, mostly digital — $6 beats $180.

A $6 placement with 0.4 NBRx of uplift beats a $180 call with 0.45.

05

Evaluating a policy: OPE and bandits

Score the new policy on the old log before anyone changes the call plan. Then keep learning. Notebook §8–9.

Offline policy evaluation — re-weight the log

Each logged row is re-weighted by π(a|x) / p(a|x): how much more (or less) the new policy would have taken that action

Logged row
context x · action a
p(a|x) · reward r

→

New policy says
π(a|x)

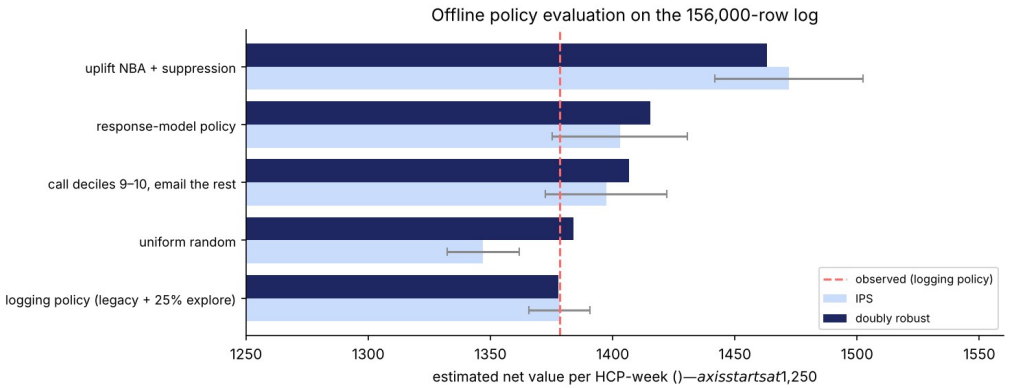
→

Weight
w = π(a|x) / p(a|x)

→

Estimate
$\hat{V}$ = mean(w · r)

Policy	IPS	SNIPS	DR	eff. n
Logging policy (sanity check)	$1,378	$1,378	$1,378	156,000
Response-model policy	$1,403	$1,411	$1,415	30,400
Uplift NBA + rules	$1,472	$1,483	$1,463	37,000



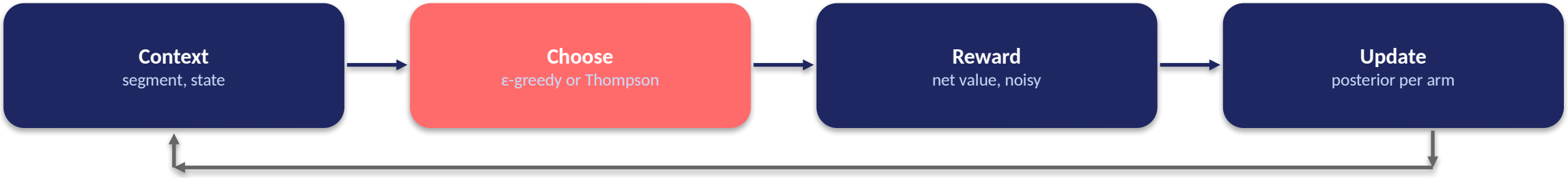
Three estimators

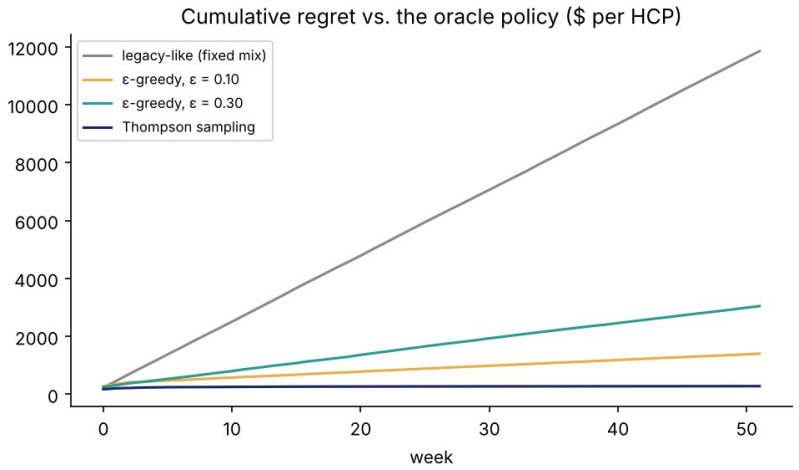
- IPS unbiased, high variance · SNIPS normalizes the weights · DR adds a reward model and is right if either part is

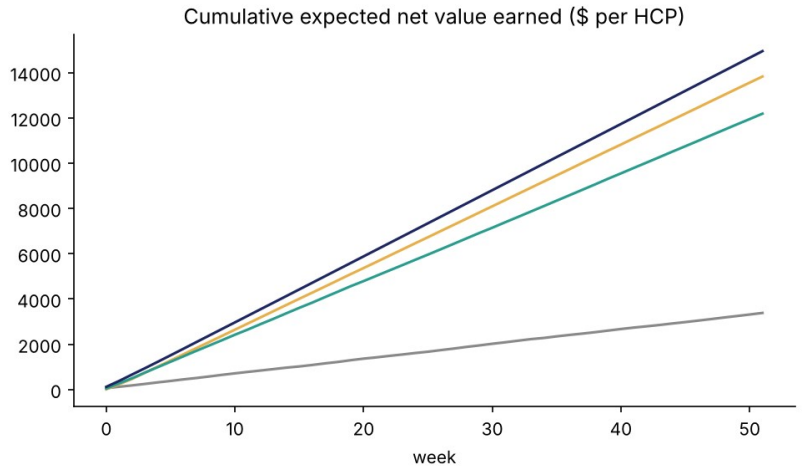
NBA: +$94 per HCP-week (±$32), all three estimators agree. Enough for an A/B test — not for a rollout.

Keep learning: the contextual bandit loop

Online-advertising vocabulary (Li et al. 2010) — not a pharma term, but the mechanism behind 'exploration'







Policy (52 wks)	Regret / HCP
Fixed mix	$11,848
ε = 0.10	$1,403
ε = 0.30	$3,044
Thompson	$282

▸ Thompson explores where uncertain, then stops.

The exploration arm of §1 is the same thing seen from the other side.

06

Patients, orchestration, exercise

The same engine for patients, under stricter governance; what turns a model into a program. Notebook §10–11.

The patient-side engine

Week 4 patient file · discontinuation within 12 months · AUC 0.77

Patient journey
claims → PDC, gaps, copay, PSP, portal

→

Risk score
GBM, allowed features only

→

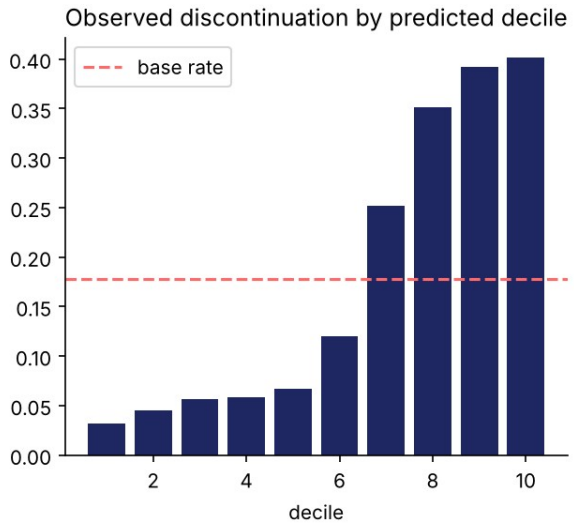
Rules
consent · purpose · fairness by payer

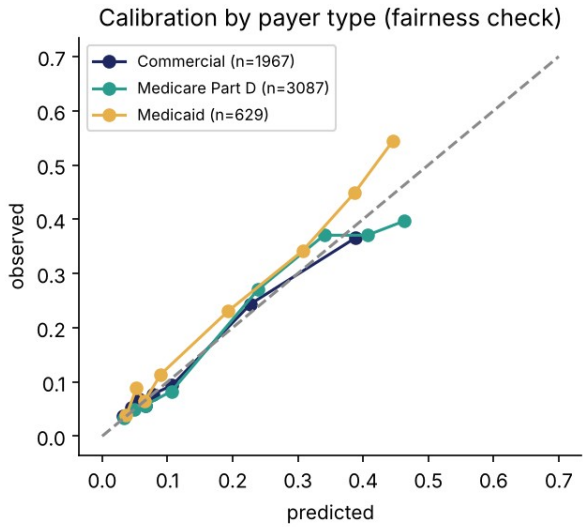
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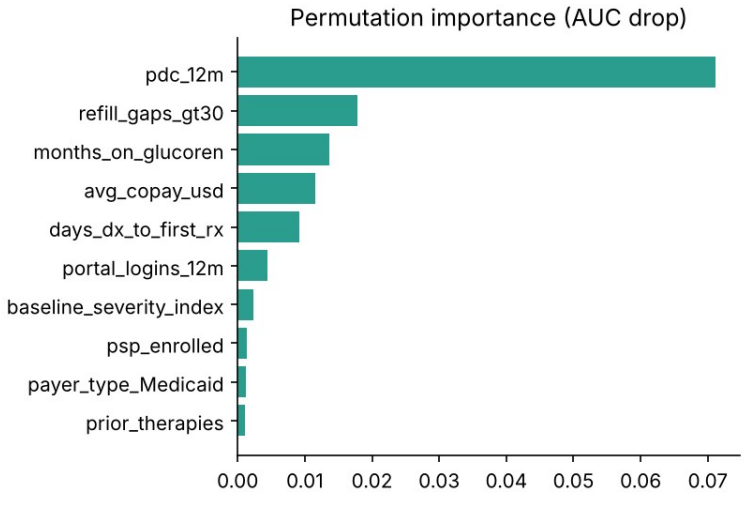
Outreach
PSP enrollment · copay card · nudge

→

Nurse decides
model ranks, human acts







Top two risk deciles: 40% discontinue (base 18%) and hold 45% of all discontinuers. That is a nurse's call list.

6 · GOVERNANCE

A score about a person

Six controls, in the order a privacy review asks

Purpose limitation
supports adherence; never deprioritizes

Data minimization
the feature list, nothing else

Channel consent
PSP consent; HCP office as default

Fairness before launch
calibration by payer and specialty

Drift monitoring
monthly, with an owner

Human in the loop
nurse decides; override rate is a metric

IQVIA, Trustworthy AI/ML for Patient Analytics: outputs such as "risk stratification scores" undergo validation; metrics include "fairness and bias detection" and drift monitoring.

6 · ORCHESTRATION

What turns a model into a program

IQVIA: a 'common data spine'. ZS/Axtria: next best experience, closed-loop learning.

Measurement
10% global holdout · 25% exploration · OPE quarterly · A/B of engine versions · report incremental NBRx and net value, not engagement

Activation
CRM suggestion to the rep · email platform · digital · patient support program

Decision
segmentation → affinity prediction → business rules → uplift model → selection + exploration

Data (CDP)
identity · consent · touch history · Rx · segments — the single customer view

Triggers

- NBRx jump · reject rate > 25% · Week 4 exit trigger · refill gap > 30 days

Owners

- Rules: compliance + field · content: access + medical · model: analytics · rollout: brand

6 · PITFALLS

Five ways NBA programs fail

Optimizing engagement
opens and clicks drift the engine toward Core

Turning off exploration
by Q3 nobody can tell Movers became Core

A response model with a causal name
'propensity to prescribe' on observational rows = §3

Rules nobody can read
estimate the cap, then write it down

Rolling out on the OPE
it justifies the A/B, not the launch

When not to build one
< 1,500 HCPs · nothing can be randomized · the constraint is access, not attention

6 · EXERCISE

In-class exercise — 20 minutes, groups of 3–4

Week_05_InClass_Exercise.docx

A · Uplift by hand (7 min)

- Eight HCPs from the randomized arm. Response, uplift, net value per action.
- Who would a response model call? Who does the engine touch?

B · Write the rules (8 min)

- Per Week 4 segment: default action, cap, one rule, one trigger.
- Which segment does the engine leave alone?

C · Pick the evaluation (5 min)

- Six scenarios: OPE, global holdout, A/B, geo test, or 'cannot be evaluated'.

Short on time: Part A only. Answer key in the instructor folder.

6 · NEXT WEEK

Week 6 — Forecasting

Monday, October 19 · Homework 2 released

Before Week 6

- **Read:** Reading Pack Parts B–D; Athey & Imbens §1–3 (skim proofs).
- **Watch:** PMSA 'From Multichannel to Omnichannel', part 1.
- **Install:** statsforecast.
- **Submit:** Homework 1 — tonight, 11:59 PM ET.

Bring

- **Your Part B rule table.**
- **One answer:** what share of weeks would your brand team accept as randomized, and what would you offer for it?
- **Term project:** segmentation + NBA archetype → the exploration arm goes in your data plan, or your evaluation is observational and says so.

Office hours Mondays 5:00–6:45 PM · ac5562@columbia.edu

Questions?

1 · Model the difference, not the level.
2 · Randomize a share of decisions — and keep it on.
3 · Segmentation says who; affinity (collaborative filtering) says how to reach them; uplift says whether it pays.
4 · The marginal touch has its own curve; the rules run in front of the model.
5 · Score offline, quote the interval, then A/B.
6 · Patients: same engine, stricter governance, a human at the point of contact.

Next week — Forecasting · Homework 2 released October 19